

A: Steganography

Descriere

Greetings 007!

Once again, Lyutsifer Safin is up to no good. We know this, as our intelligence services have been able to intercept two images that have been circulating in the underground world. The images only differ slightly and so, we believe that `lyutsifer-ig.pgm` contains a hidden message that Safin is spreading around. We believe the message was encoded using the LSB Steganography algorithm.

Your mission, should you choose to accept it, is to find out the hidden message and stop Safin before it's too late. As you know, Safin always ends his convos with `LSSL`. Maybe it will come in handy during this task.

Good luck extracting the message!

Background

Steganography is the practice of concealing a message within another message or a physical object. In computing/electronic contexts, a computer file, message, image, or video is concealed within another file, message, image, or video.

LSB Steganography is an image steganography technique in which messages are hidden inside an image by replacing each pixel's least significant two bits with the bits of the message to be hidden.

A PGM digital image is represented as a matrix of pixels whose values range from 0 to 255. Let's assume that the values of the first four pixels are 92 81 103 144, and we want to conceal the character 'a' inside them.

The binary representation of the pixels is 01011100 01010001 01100111 10010000 and the binary representation of the character a is 01100001

To conceal the character, we will take the first two bits 01 of the character (considering the direction from left to right, also known as Most Significant Bits - MSB) and store them in the last two bits (LSB) of the first pixel. Thus, the pixel 01011100 becomes 01011101. We then take the next two bits of the char, 10, and store them in the next pixel. And so on.

At the end, our pixels will have the binary values 01011101 01010010 01100100 10010001.

Doing so, we can hide a character using four pixels. Because we're modifying only the last two bits, the differences between the originals and the encoded pixels will be virtually indistinguishable by the human eye.

Download the source file [here](#).

The original image is available [here](#).

The intercepted image is available [here](#).

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Make sure that your solution also outputs Safin's signature!

Exemple